

Evaluating Machine Learning Algorithms for Predicting Tuberculosis Cases in Mulago National Referral hospital: A Comparative Analysis of Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting Models

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Abstract

Background: Tuberculosis (TB) remains a critical public health crisis in Uganda, with approximately 22% of cases; roughly 10,000 annually remaining undetected. At Mulago National Referral Hospital, high patient volumes and late-stage symptomatic screening create significant diagnostic bottlenecks. This study evaluated four supervised machine learning (ML) architectures to determine the most stable and interpretable framework for early TB case prediction in resource-constrained tertiary settings.

Methods: We analyzed a cross-sectional dataset of 529 patient records from the Mulago TB Unit Register (FY 2023/24). A rigorous 70/15/15 split was employed for training, validation, and testing. We compared Logistic Regression (LR), Decision Trees (DT), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Models were optimized via hyperparameter and evaluated using a multi-metric approach; Accuracy, Precision, Recall (Sensitivity), F1-score, and Area Under the ROC Curve (AUC). A Chi-square Analysis and Recursive Feature Elimination (RFE) was used to rank predictors significantly.

Results: In the testing phase, XGBoost achieved the highest predictive power (F1-score: 88%; AUC: 92%). Followed by Random Forest (F1-score: 82%; AUC: 92%). However, upon exposure to unseen validation data, Logistic Regression emerged as the most stable and generalizable model, maintaining an F1-score of 78% and an AUC of 88%. Across all models, the primary predictors for TB infection were identified as Male Sex (AOR = 1.45; 95% CI: 1.17-1.80) and Urban Residence (AOR = 1.43; 95% CI: 1.15-1.77). The highest-risk clinical profile consisted of HIV-positive males in urban settings who had previously completed treatment without a confirmed cure.

Conclusions: While ensemble methods like XGBoost offer superior raw performance, Logistic Regression provides the optimal balance of stability and clinical interpretability required for hospital-level integration. Implementing such ML-driven risk-stratification tools within Uganda's electronic health records could significantly enhance early case detection and optimize the allocation of GeneXpert and radiography resources.

Keywords: Tuberculosis, Machine Learning, Predictive Modeling, Logistic Regression, Random Forest, XGBoost, Uganda.

1. Introduction.

The Global and National Burden of Tuberculosis.

Tuberculosis (TB) remains one of the most significant threats to global public health, claiming approximately 1.3 million lives in 2023 alone (WHO, 2024). Uganda is categorized by the World Health Organization (WHO) as one of the 30 high-burden countries for TB and TB/HIV co-infection (WHO, 2021). Despite aggressive national interventions, the Uganda Ministry of Health (MOH) reported that nearly 22% of cases equivalent to 10,000 infections go undetected annually (WHO, 2025). This diagnostic gap facilitates continued community transmission and elevates mortality rates, particularly in urban “hotspots” like Kampala.

The Diagnostic Bottleneck at Mulago National Referral Hospital.

As the primary tertiary healthcare facility in Uganda, Mulago National Referral Hospital manages an overwhelming volume of presumptive TB cases. However, the reliance on traditional diagnostic pathways primarily sputum smear microscopy and GeneXpert is highly specific, its high cost, requirement for stable power, and turnaround times limit its utility for mass initial screening in high-traffic settings (WHO., 2014, Nalugwa et al., 2020). Consequently, many patients are only screened after presenting with advance symptoms, missing the window for early intervention (Muzeyi et al., 2025, Buregyeya et al., 2014).

The Limitations of Traditional Statistical Modeling.

Historically, clinical prediction has relied on Logistic Regression (LR) to identify risk factors (Hosmer Jr et al., 2013). While LR offers excellent interpretability and remains a “gold standard” in biostatistics, it often struggles to capture the complex, non-linear relationships and high-dimensional interactions inherent in multifaceted medical datasets (Sidey-Gibbons and Sidey-Gibbons, 2019). The rise of Machine Learning (ML) offers a paradigm shift; ensemble methods like Random Forest (RF) and Extreme Gradient Boosting (XGBoost) can process heterogeneous clinical variables more effectively, potentially offering higher sensitivity in identifying “hidden” TB cases (Babirye et al., 2024).

The Study and its Significance.

There is currently a shortage of localized evidence comparing advanced ML architectures against statistical models within the Ugandan referral context. This study addresses the gap by evaluating and comparing the performance of four supervised ML algorithms; Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting, using routine clinical and demographic data from the Mulago TB Unit (FY 2023/24). By identifying the most stable and accurate predictive model, this research aims to provide a data-driven tool for clinicians to triage high-risk patients, ensuring that limited diagnostic resources like GeneXpert are allocated to those with the highest probability of infection.

2. Methods.

Study Design and Setting.

The study employed a retrospective cross-sectional study design utilizing routine clinical data from the Mulago National Referral Hospital TB Unit. Mulago serves as the primary referral center for the Kampala metropolitan area and surrounding districts, managing a high density of presumptive TB cases. Data were extracted from the integrated TB registers covering the 2023/2024 financial year.

Data Source and Study Population.

The initial dataset consisted of 542 entries. Following data cleaning, 13 records were excluded due to missing primary outcome variables; that is, those with missing HIV results and whose age was below 18, resulting in a final analytical sample of 529 observations. The target variable was “TB Status” a binary outcome defined by laboratory-confirmed results (GeneXpert or Smear Microscopy)

Feature Engineering and Selection.

All Seventeen independent variables were initially considered, including demographic (Age, Sex, Residence), clinical (HIV status, History of TB, weight) and other domains. To mitigate the risk of data leakage, variables directly resulting from a TB diagnosis (treatment regimens) were excluded during the feature engineering phase.

We utilized Recursive Feature Elimination (RFE) with a logistic regression estimator to identify the most parsimonious set of predictors. This process prioritized features that contributed most significantly to the model’s refractive power, ultimately focusing on 10 predictors for the final architectures.

Machine Learning framework and Partitioning.

The dataset was partitioned using a 70/15/15 split;

- Training set (70%): Used for initial model fitting.
- Testing set (15%): Used for hyperparameter optimization and to guard against overfitting.
- Validation set (15%): Reserved as hold-out data to evaluate final model generalizability.

Model Architectures and optimization.

Four supervised learning algorithms were developed and compared;

1. Logistic Regression (LR): Employed as the baseline statistical model, utilizing a logit link function to eliminate Adjusted Odds Ratios (AORs).
2. Decision Trees (DT): Implemented using the CART algorithm to visualize hierarchical decision rules.
3. Random Forest (RF): An ensemble of 500 bootstrap-aggregated trees to reduce variance.

4. XGBoost (Extreme Gradient Boosting): A gradient-boosted framework optimized through fine-tuning of the learning rate (eta), maximum depth, and subsample ratio to maximize the F1-score.

Evaluation Metrics.

Given the public health implications of undetected TB, the models were primarily evaluated on Recall (Sensitivity) and the F1-score (the harmonic mean of precision and recall). We also generated Receiver Operating Characteristic (ROC) curves to calculate the Area Under the Curve (AUC), providing a threshold-independent measure of classifier performance.

Ethical Considerations.

The study was approved by the Makerere University School of Public Health Research and Ethics Committee. Data were fully anonymized prior to analysis to ensure patient confidentiality.

3. Results

Study Population and Descriptive Statistics

The final analysis included 529 patient records from the Mulago TB Unit. The cohort was predominantly male (63.3%), with a mean age of 38.4 years ($SD \pm 12.2$). A significant proportion of the study population resided in urban areas (66.7%), and the HIV prevalence among presumptive TB cases was 59.7%. Laboratory confirmation (GeneXpert/Smear) identified a TB positivity rate of 42.1% within the sample.

Variable Importance and Predictor Ranking

A Chi-square Analysis was conducted first to determine what variables were statistically significant. This was later followed by the Recursive Feature Elimination (RFE) method as a confirmatory machine learning test. Recursive Feature Elimination (RFE) identified Sex, Residence, and HIV Status as the most influential predictors across all machine learning architectures. The variables that were significant were included into the final model diagnostics as shown in table 1 below, the variables that were excluded were; those that were not statistically significant and those that had 100% representation because they carry no information in explaining the variations in the outcome. Notably, the “Treatment Outcome” variable showed high predictive power, specifically among patients who had previously completed treatment without a confirmed cure.

A bivariate analysis of independent variables by TB status and Feature importance ranking

Variables	Freq (%)	Tb negative (%)	Tb positive (%)	Sig	RFE Rank
HIV status					

Negative	213 (40.3)	84 (39.4)	129 (60.6)	>0.001***	1
Positive	316 (59.7)	221 (69.9)	95 (30.1)		
Age					
18-20	21 (4.0)	8 (38.1)	13 (61.9)	0.017*	2
21-30	146 (27.6)	78 (53.4)	68 (46.6)		
31-40	172 (32.5)	94 (54.7)	78 (45.3)		
40+	190 (35.9)	125 (65.8)	65 (34.2)		
Sex					
Female	194 (36.7)	134 (69.1)	60 (30.9)	>0.001***	1
Male	335 (63.3)	171 (51.0)	164 (49.0)		
Weight category (kg)					
Normal weight (51-70)	228 (43.1)	127 (55.7)	101 (44.3)	0.6728	3
Under weight (<=50)	269 (50.9)	158 (58.7)	111 (41.3)		
Over weight (> 70)	32 (6.0)	20 (62.5)	12 (37.5)		
Type of Patient					
New	519 (98.1)	299	220 (42.4)	1	6
Others	10 (1.9)	6 (60.0)	4 (40.0)		
History of TB					
No	519 (98.1)	299 (57.6)	220 (42.4)	1	5
Yes	10 1.9	6 (60.0)	4 (40.0)		
Source of referral					
Community	16 (3.0)	7 (43.8)	9 (56.3)	0.3755	4
Facility	513 (97.0)	298 (58.1)	215 (41.9)		
Place of Residence					
Rural	176 (33.3)	127 (72.2)	49 (27.8)	>0.001***	1
Urban	353 (66.7)	178 (50.4)	175 (49.6)		
Disease Classification					
Extra pulmonary (EP) TB	76 (14.4)	72 (94.7)	4 (5.3)	>0.001***	1
Pulmonary Clinically Diagnosed (P-CD) TB	113 (21.4)	112 99.1	1 (0.9)		
Pulmonary bacteriologically confirmed (P-BC) TB	340 (64.3)	121 (35.6)	219 (64.4)		
Treatment outcome					
Completed treatment	321 (60.7)	138 (92.0)	12 (8.0)	>0.001***	1
Cured	58 (11.0)	123 (38.3)	198 (61.7)		
Died/Lost to Follow up/Not evaluated	150 (28.4)	44 (75.9)	14 (24.1)		
Patient category					

National	529 (100.0)	305 (57.7%)	224 (42.3%)	>0.001***	7
Diagnosed with DR_TB					
No	529 (100.0)	305 (57.7%)	224 (42.3%)	>0.001***	7
Treatment Model					
Non-Digital community DOT	529 (100.0)	305 (57.7%)	224 (42.3%)	>0.001***	7
<i>p-value <0.001 (***) <0.01 (**) < 0.05 (*)</i>					

Comparative Performance of Machine Learning Models.

The performance of the four algorithms was evaluated across two stages; initial testing (hold-out set) and final validation (unseen data).

- XGBoost: In the testing phase, XGBoost demonstrated the highest raw predictive power, achieving an F1-score of 88% and an AUC of 92%. It showed a superior ability to minimize false negative (Recall: 89%), which is critical for TB screening.
- Random Forest: Performed strongly with an F1-score of 82% and an AUC of 92%, showing high robustness but slightly lower sensitivity than XGBoost.
- Decision Tree: While highly interpretable, the DT model showed evidence of overfitting, with a significant drop in performance when moving from training to testing (F1-score: 74%)

The Stability-Accuracy Trade-off.

A critical finding of this study was the performance of Logistic Regression (LR) during the validation phase. While XGBoost had higher peaks, LR maintained the most consistent performance on unseen data, with a validation F1-score of 78% and an Accuracy of 78%. This suggests that for the current sample size at Mulago, the simpler linear model may offer better generalizability across different patient cohorts.

A table showing a Comparative Performance of ML Model on the Validation dataset.

ML Model	Performance Metrics	Validation	Final Average Validation
Logistic Regression	Precision	78%	78%
	Recall	78%	
	F1-Score	78%	
Decision Tree	Precision	73%	73%
	Recall	74%	
	F1-Score	72%	
Random Forest	Precision	77%	76%
	Recall	76%	
	F1-Score	76%	
XGBoost	Precision	75%	

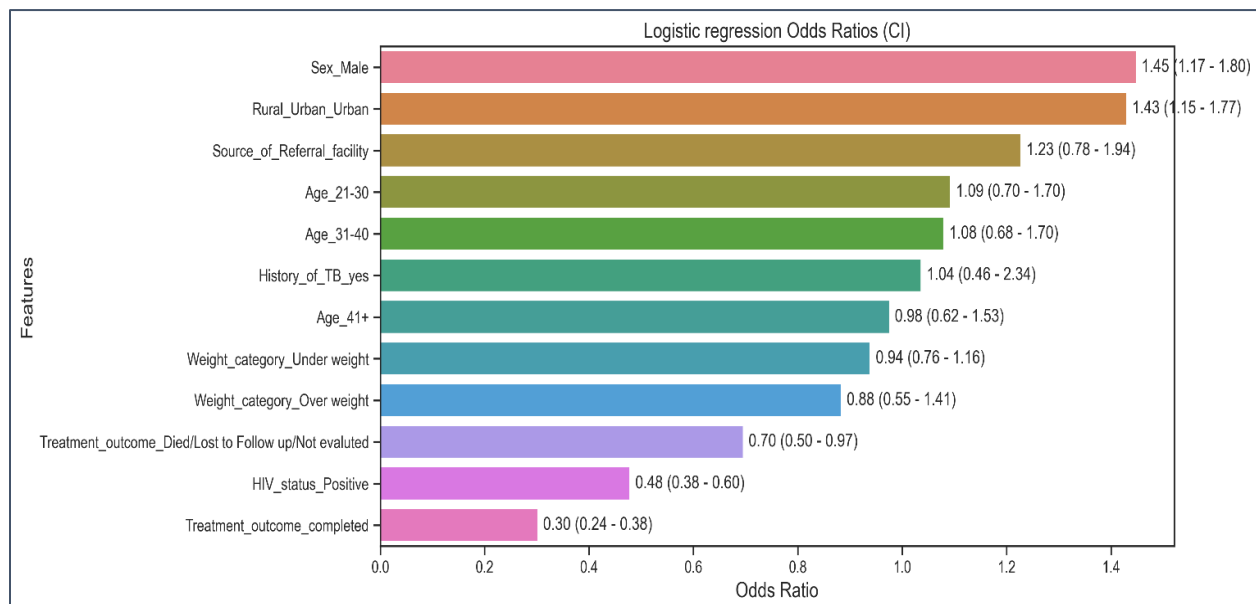
Recall	75%	75%
F1-Score	75%	

Risk Factor Analysis (Adjusted Odds Ratio).

Multivariable analysis using the best-performing stable model (Logistic Regression) revealed that;

- Males were 1.45 times more likely to contract TB compared to the women (AOR = 1.45; 95% CI: 1.17-1.80), controlling for other factors.
- Respondents who resided in Urban areas were 1.43 times more likely to contract TB compared to those who reside in Rural areas (AOR = 1.43; 95% CI: 1.15-1.77), controlling for other factors.

A graph showing adjusted odds ratios (AOR) ranked in ascending order as identified by the Logistic Regression.



4. Discussion

Comparative performance and the Complexity Paradox.

The primary objective of this study was to compare different machine learning models to determine applicability to TB screening prediction, with the aim of identifying the key predictors influencing TB disease and the most effective model in supporting early detection. Our results demonstrate a complexity paradox; while XGBoost and Random Forest achieved superior predictive power in controlled testing (AUC: 0.92), Logistic Regression proved to be the more stable and generalizable

model during validation (78% F1-score) (Sidey-Gibbons and Sidey-Gibbons, 2019, Chen et al., 2024).

This divergence is likely attributable to the Bias-Variance Trade-off. Ensemble models like XGBoost are highly sensitive to complex patterns but can easily overfit smaller, high-dimensional datasets common in hospital registers (Riley et al., 2020). In contrast, the linear constraints of Logistic Regression provided a regularizing effect, making it more robust to the inherent noise in routine clinical data at Mulago (Sidey-Gibbons and Sidey-Gibbons, 2019).

Clinical Significance of Identified Predictors.

The models identified Male Sex and Urban Residence as the strongest predictors for TB positivity. The finding that males are 1.45 times more likely to be infected aligns with recent epidemiological trends in Sub-Saharan Africa, where men often delay health-seeking behavior and have higher exposure in communal work environments (WHO, 2024, Muttamba et al., 2024).

Furthermore, the significant association with Urban Residence (AOR 1.43) highlights the impact of overcrowding and poor ventilation in Kampala's informal settlements, which acts as catalysts for TB transmission (Aturinde et al., 2025). Most critically, the high predictive weight assigned to previous treatment completion suggests a potential gap in the standard of care post-treatment, where patients may be cleared without biological confirmation of cure, leading to relapse or undetected persistence (Omara et al., 2025).

Practical Implications for Mulago National Referral Hospital.

From a public health perspective, the cost of false negative (missing a TB case) far outweighs the cost of a false positive (unnecessary GeneXpert testing). Therefore, the high Recall (Sensitivity) of our models is their most valuable feature.

This study proposes a two-tiered triage system;

- Tier 1: A simple, LR-based digital screening tool integrated into the Mulago electronic records to flag high-risk individuals based on demographic and historical data.
- Tier 2: Prioritizing these high-risk individuals for immediate GeneXpert MTB/RIF and digital chest radiography, thereby optimizing the allocation of expensive diagnostic reagents.

Limitations.

While robust, this study is limited by its retrospective nature and reliance on hospital registers, which may be subject to entry errors (Nalugwa et al., 2020). Additionally, the sample size (n=529), while sufficient for statistical significance, may limit the full potential of deep learning or more complex neural architectures which require big data to reach peak efficiency (Sidey-Gibbons and Sidey-Gibbons, 2019, Chen et al., 2024).

5. Conclusion

This study demonstrates that machine learning algorithms, particularly XGBoost and Logistic Regression, offer a robust framework for predicting Tuberculosis cases at Mulago National Referral Hospital. While XGBoost provides superior sensitivity, the stability and interpretability of Logistic Regression make it the most viable candidate for immediate integration into hospital-based electronic decision-support systems.

By leveraging routinely collected demographic and clinical data specifically targeting high-risk urban males and HIV-positive individuals, clinicians can significantly improve early case detection. We recommend the adoption of these predictive models as a primary triage tool to optimize the use of high-cost diagnostic resources like GeneXpert, ultimately reducing the burden of undetected TB in Uganda.

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